Date

OL = Outlook, T = Temperature, H = Humidity, W = Windy, R = Rain

Predict rain

OL	T	H	W	R
ny	Hot	High	Weak	No
nny	Mild	High	Strong	No
loudy	Hot	High	Weak	Yes
ain	Mild	High	Weak	Yes
ain	~~Hot~~Cool	Normal	Weak	Yes
ain	Cool	Normal	Strong	No
oudy	Mild	Normal	Strong	Yes
nny	Cool	Normal	Weak	No

$$\text{Total Entropy} = -\left[P(\text{Yes}) \log_2 (P(\text{Yes}))\right] - \left[P(\text{No}) \log_2 (P(\text{No}))\right]$$

$$= -\left[\frac{4}{8} \times \log_2 \left(\frac{4}{8}\right)\right] - \left[\frac{4}{8} \log_2 \left(\frac{4}{8}\right)\right]$$

~~Total En = 0.5 1~~

Total Entropy = 1

Date

Entropy of Outlook

	Y	N	Entropy
Sunny	0	3	0
Cloudy	2	0	0
Rainy	2	1	0.918

Sunny Entropy $= -\left[\left(\frac{0}{3}\right) \times \log_2\left(\frac{0}{3}\right)\right] - \left[\left(\frac{3}{3}\right) \times \log_2\left(\frac{3}{3}\right)\right] = 0$

Cloudy Entropy $= -\left[\left(\frac{2}{2}\right) \times \log_2\left(\frac{2}{2}\right)\right] - \left[\left(\frac{0}{2}\right) \log_2\left(\frac{0}{2}\right)\right]$

Rainy Entropy $= -\left[\left(\frac{2}{3}\right) \log_2\left(\frac{2}{3}\right)\right] - \left[\left(\frac{1}{3}\right) \log_2\left(\frac{1}{3}\right)\right]$

$= 0.918$

Weighted Entropy $= 0 + 0 + \left(\frac{3}{8} \times 0.918\right)$

Weighted Entropy of Outlook $= 0.344$

Info gain of Outlook $= 1 - 0.344 = \boxed{0.656}$

Date

Entropy of Temperature

	Y	N	Entropy
Hot	1	1	1
Mild	2	1	0.779
Cool	1	2	0.918

Hot Entropy $= -\left[\left(\frac{1}{2}\right)\log_2\left(\frac{1}{2}\right)\right] - \left[\left(\frac{1}{2}\right)\log_2\left(\frac{1}{2}\right)\right] = 1$

Mild Entropy $= -\left[\left(\frac{2}{3}\right)\log_2\left(\frac{2}{3}\right)\right] - \left[\left(\frac{1}{3}\right)\log_2\left(\frac{1}{3}\right)\right] = 0.779$

Cool Entropy $= -\left[\left(\frac{1}{3}\right)\log_2\left(\frac{1}{3}\right)\right] - \left[\left(\frac{2}{3}\right)\log_2\left(\frac{2}{3}\right)\right] = 0.918$

Weighted Entropy $= \left[\frac{2}{8} \times 1 = 0.25\right] + \left[\left(\frac{3}{8}\right) \times 0.779 = 0.292\right]$

$+ \left[\frac{3}{8} \times 0.918 = 0.344\right] = 0.886$

Info gain of Temperature $= 1 - 0.886 = \boxed{0.114}$

Signature_____ RC No._____

Date

Entropy of Humidity :-

	Y	N	Entropy
High	2	2	1
Normal	2	2	1

$$\text{High Entropy} = -\left[\left(\frac{2}{4}\right) \times \log_2\left(\frac{2}{4}\right)\right] - \left[\frac{2}{4} \times \log_2\left(\frac{2}{4}\right)\right]$$

$$= 1$$

$$\text{Normal Entropy} = -\left[\left(\frac{2}{4}\right) \times \log_2\left(\frac{2}{4}\right)\right] - \left[\frac{2}{4} \times \log_2\left(\frac{2}{4}\right)\right]$$

$$= 1$$

$$\text{Weighted Entropy} = \left(\frac{4}{8} \times 1 = 0.5\right) + \left(\frac{4}{8} \times 1 = 0.5\right)$$

~~Info-gain~~ $= 1$

Info gain $= 1 - 1 = \boxed{0}$

Signature

RC

No.

Date

Entropy of Windy

	Y	N	Entropy
Weak	3	2	0.970
Strong	1	2	0.918

$$\text{Weak Entropy} = -\left[\frac{3}{5}\log_2\left(\frac{3}{5}\right)\right] - \left[\frac{2}{5}\times\log_2\left(\frac{2}{5}\right)\right]$$

$$= 0.970$$

$$\text{Strong Entropy} = -\left[\frac{1}{3}\times\log_2\left(\frac{1}{3}\right)\right] - \left[\frac{2}{3}\times\log_2\left(\frac{2}{3}\right)\right]$$

$$= 0.918$$

$$\text{Weighted Entropy} = \left(\frac{5}{8}\times 0.970\right) + \left(\frac{3}{8}\times 0.918\right)$$

$$= 0.606 + 0.344 = 0.95$$

$$\text{Info gain} = 1 - 0.95 = \boxed{0.05}$$

Signature

RC

No.

Pg: 6

Date

Outlook is root node